

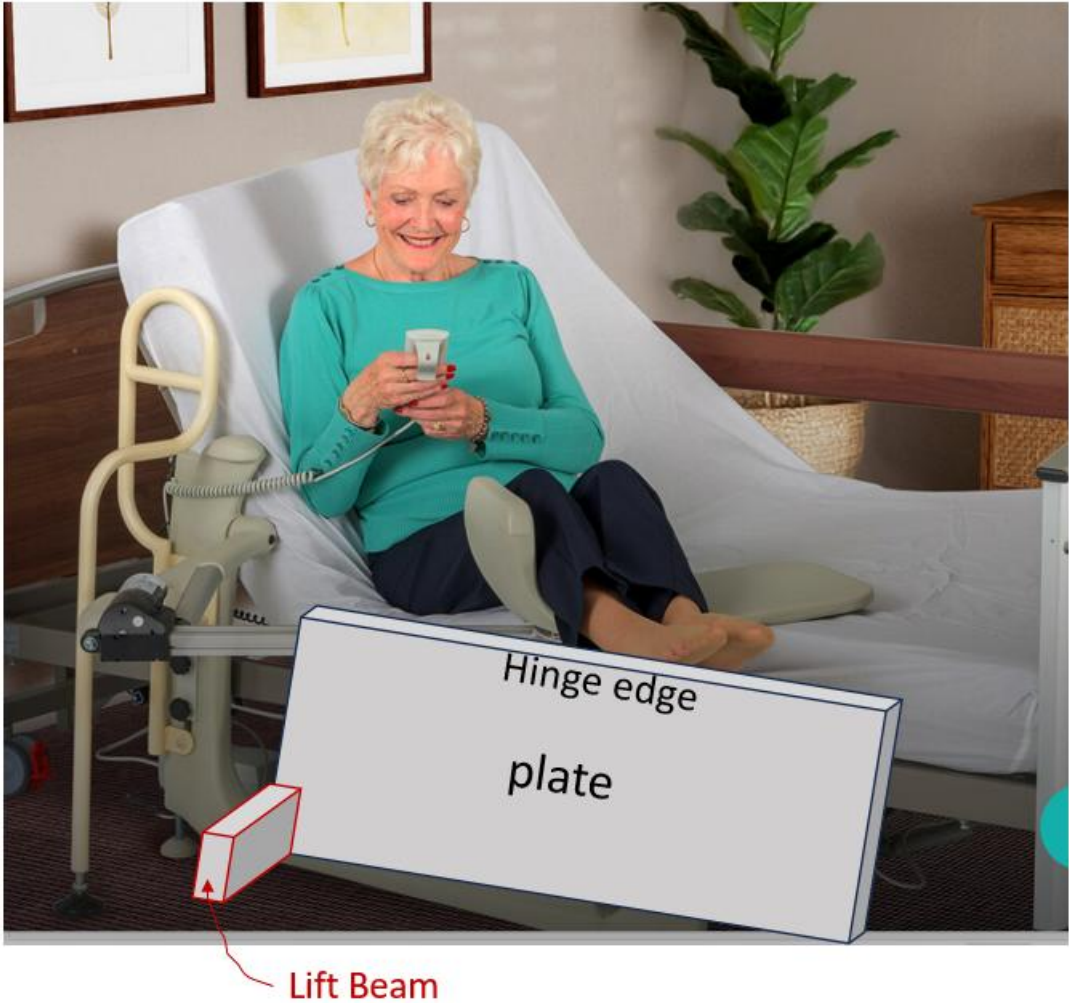
Consider the design of a rectangular plate that is a key component of an assistive device (prototype), to help a patient lift his/her legs to get into bed. The plate will be actuated using a cable attached to a beam-column fixed to the bottom right corner of the plate. A hinge on the top edge of the plate constrains the plate to rotate. The plate should not deflect too much so that the patient can easily slide her legs across the plate after it has been raised to a horizontal position. Your design goal for now is to specify the plate material and structure, and column-beam material and cross-section, to get an economical light easy to construction prototype that can carry a specified load, with a specified maximum deflection and factor of safety.

For this exercise, you can consider the applied weight to be centrally loaded at a point for simplicity. Think about how you can model the plate and beam, using mechanics of materials models that are simple and relatively easy to use. What type of loading, and what end conditions, are appropriate to model the plate? What type of loading, and what end conditions, are appropriate to model the beam? Would/could you incorporate any beams along the free edges of the plate to stiffen the plate?

Note: start all work using symbols for dimensions, loads, and properties, because in later assignments I may vary the specifications for different design groups. Consider using Excel (noting and checking your units) when you start substituting (and varying) properties and dimension values.

Given:

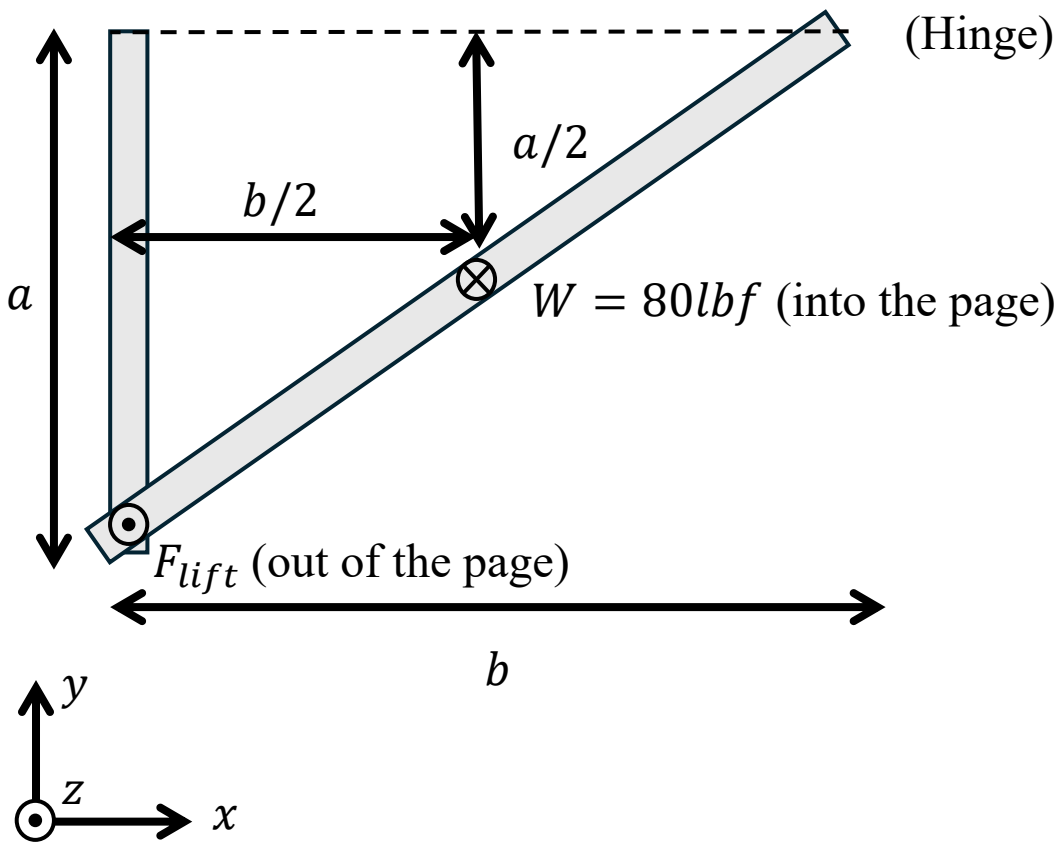
plate depth, a	17"
plate width, b	36"
plate center load, W	80 lb
Factor of Safety, FoS	2.5
Maximum Deflection, $d_{\max}$	0.5"



Create a design model for the plate, that relates maximum stresses to plate thickness, depth, width, and load, and that predicts plate maximum deflection. Be specific about the edge conditions for your model, and justify the conditions based on your design. Discuss limitations and simplifications that you had to use, and how these might limit your choice of materials.

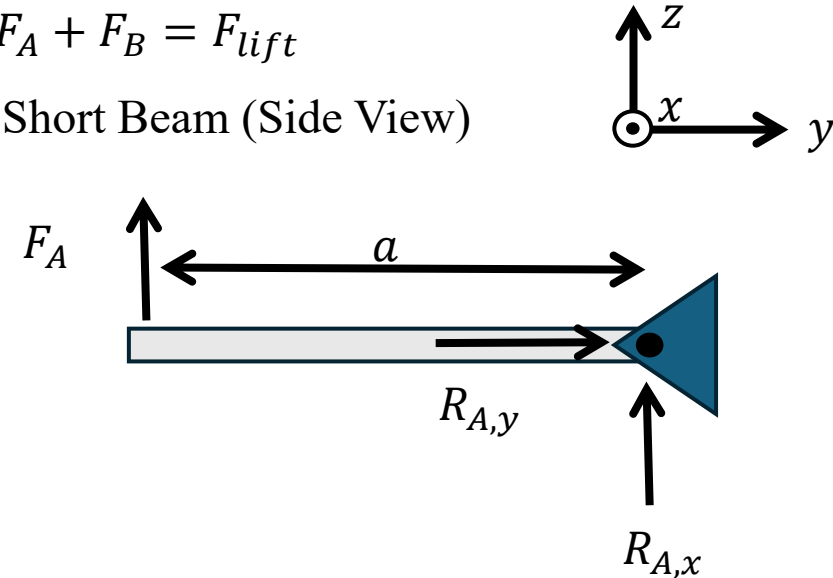
Plate Model in Fully Lifted Position

Top View

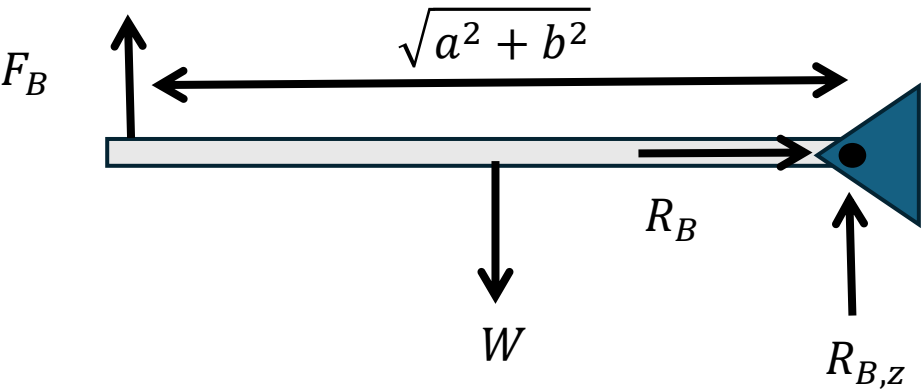


$$F_A + F_B = F_{lift}$$

Short Beam (Side View)



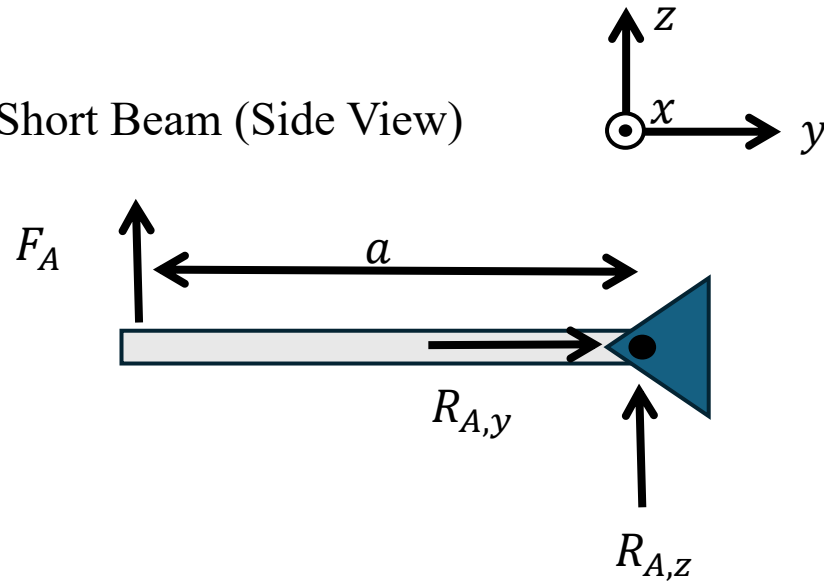
Diagonal Beam (Side View)



Statics Analysis

Constraint equation: $F_A + F_B = F_{lift}$

Short Beam (Side View)



$$\sum F_y = R_{A,y} = 0$$

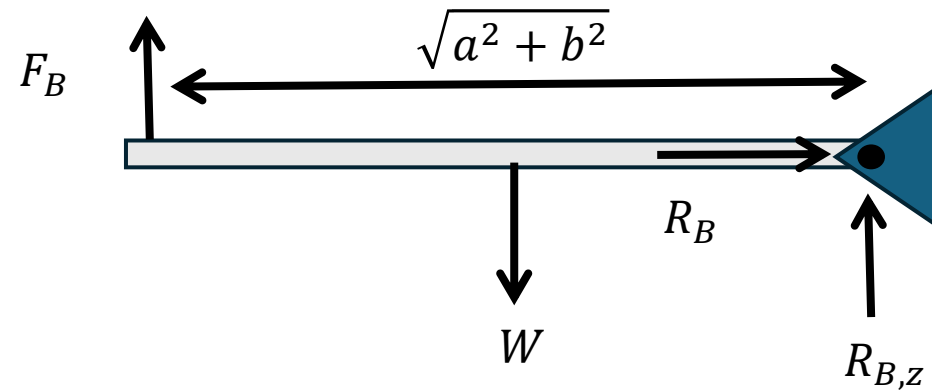
$$\sum F_z = R_{A,z} + F_A = 0 \rightarrow R_{A,z} = -F_A$$

$$\sum M_x = F_A a = 0 \rightarrow F_A = 0$$

Statics Analysis

Constraint equation: $F_A + F_B = F_{lift}$

Diagonal Beam (Side View)



Results:

- $R_{A,y} = 0$
- $R_{A,z} = 0$
- $F_A = 0$
- $R_B = 0$
- $R_{B,z} = \frac{1}{2}W = 40 \text{ lbf}$
- $F_B = \frac{1}{2}W = 40 \text{ lbf}$

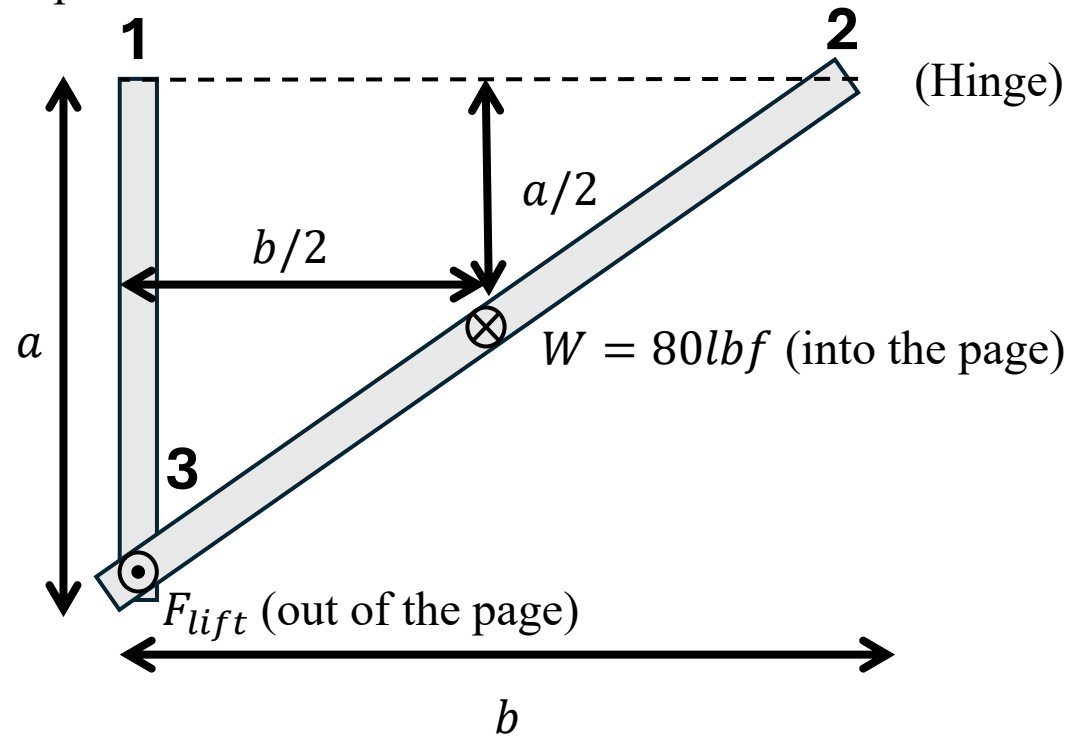
$$\sum F_{axial} = R_B = 0$$

$$\sum F_z = R_{B,z} + F_B - W = 0 \rightarrow R_{B,z} = W - F_B$$

$$\sum M = W \frac{1}{2} \sqrt{a^2 + b^2} - F_B \sqrt{a^2 + b^2} = 0 \rightarrow F_B = \frac{1}{2} W$$

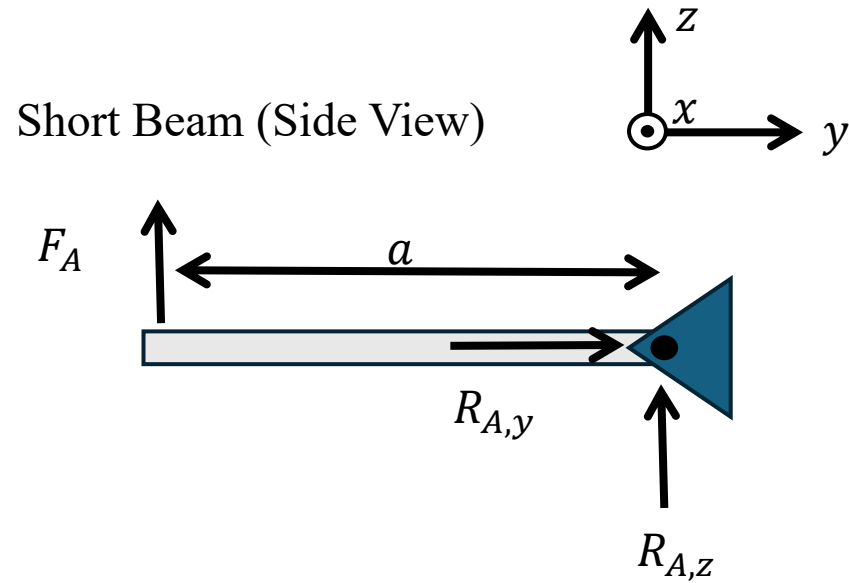
Beam Deflection Analysis**Plate Model in Fully Lifted Position**

Top View



Boundary conditions:

- The hinge prevents any deflection, thus
$$\delta_{short,1} = 0, \delta_{diagonal,2} = 0$$
- The beams are joined at 3, thus
$$\delta_{short,3} = \delta_{diagonal,3}$$

Beam Deflection Analysis

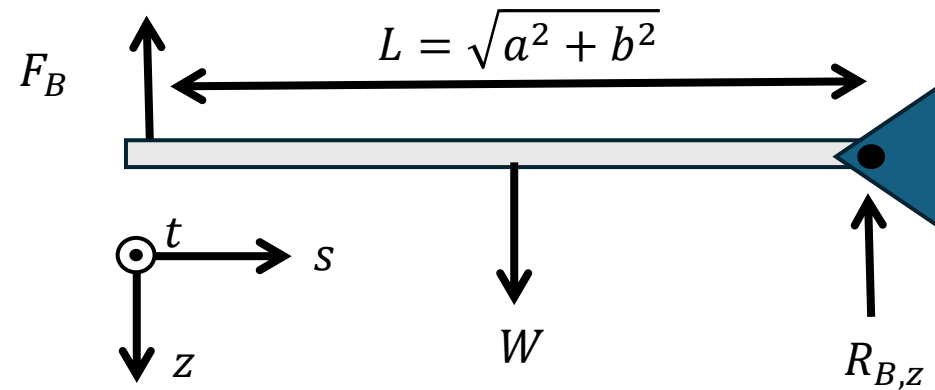
Since

- $R_{A,y} = 0$
- $R_{A,z} = 0$
- $F_A = 0$

The deflection of the short beam would be zero. Additionally, this also means the deflection of the diagonal beam end would also be zero. $\delta_{short,3} = \delta_{diagonal,3} = 0$

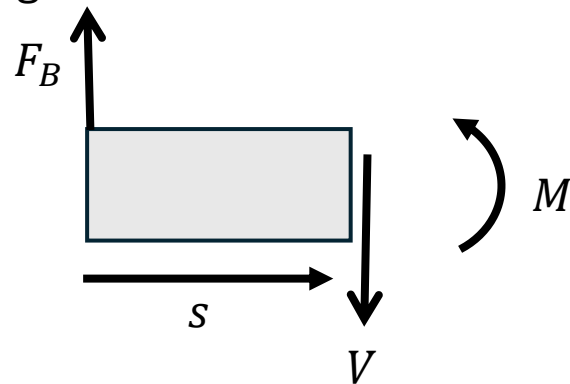
Beam Deflection Analysis

Diagonal Beam (Side View)



- $R_{B,z} = \frac{1}{2}W = 40 \text{ lbf}$
- $F_B = \frac{1}{2}W = 40 \text{ lbf}$
- $L = \sqrt{a^2 + b^2} = 39.812 \text{ in}$
- s is along the axial direction of the diagonal beam
- t is normal to the s - z plane

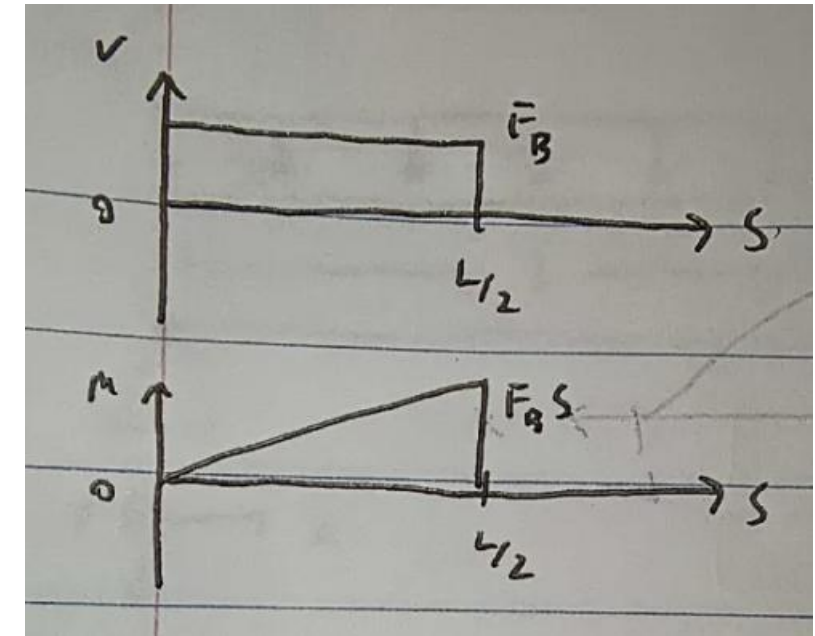
Solving for the shear and moment distribution for $(0 \leq s \leq \frac{L}{2})$:



$$\sum F_s = 0$$

$$\sum F_z = V - F_B = 0 \rightarrow V = F_B$$

$$\sum M_t = M - V s \rightarrow M = F_B s$$



Beam Deflection Analysis

Double integration for the beam deflection ($0 \leq s \leq \frac{L}{2}$):

$$\frac{d^2 z}{ds^2} = -\frac{M(s)}{EI} = -\frac{F_B}{EI} s$$

$$z(s) = -\frac{F_B}{EI} \int \int s \, ds \, ds = -\frac{F_B}{6EI} s^3 + c_1 s + c_2$$

The slope at $s = L/2$ is zero because of symmetry (the beam sags down with zero deflection about its ends):

$$z'(s) = -\frac{F_B}{2EI} s^2 + c_1$$

$$z'\left(\frac{L}{2}\right) = -\frac{F_B}{2EI} \left(\frac{L}{2}\right)^2 + c_1 = 0$$

$$c_1 = \frac{F_B L^2}{8EI}$$

The deflection is zero at the end

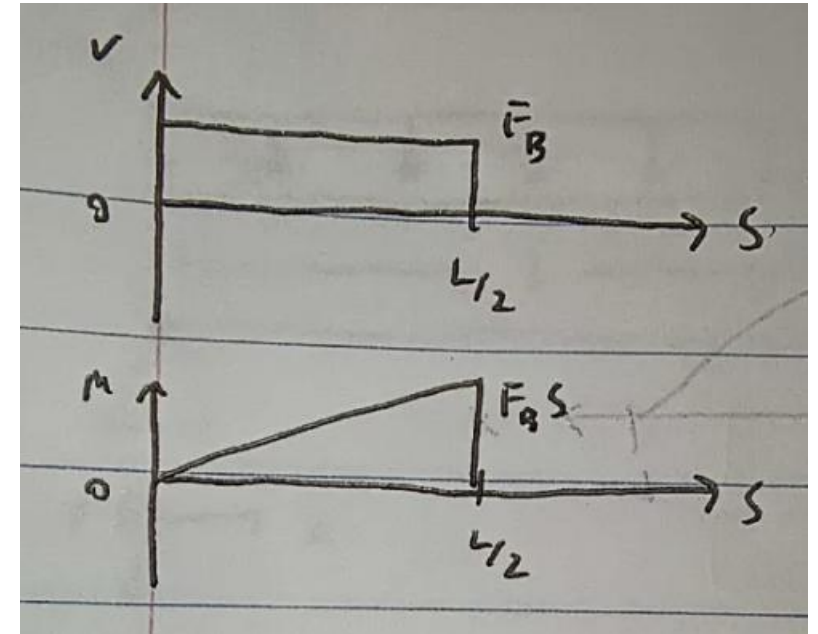
$$z(0) = c_2 = 0$$

Thus, the deflection is given by:

$$z(s) = -\frac{F_B}{6EI} s^3 + \frac{F_B L^2}{8EI} s$$

Where:

- $F_B = \frac{1}{2} W = 40 \text{ lbf}$
- $L = \sqrt{a^2 + b^2} = 39.812 \text{ in}$



Beam Deflection Analysis

The max deflection at $s = L/2$ is:

$$z(s) = -\frac{F_B}{6EI}s^3 + \frac{F_B L^2}{8EI}s = -\frac{W}{12EI}s^3 + \frac{WL^2}{16EI}s = -\frac{W}{48EI}(4s^3 - 3L^2s)$$

$$z(L/2) = -\frac{W}{48EI}\left(4\left(\frac{L}{2}\right)^3 - 3L^2\left(\frac{L}{2}\right)\right) = -\frac{W}{48EI}\left(\frac{L^3}{2} - \frac{3L^3}{2}\right)$$

$$z_{max} = \frac{W}{48EI}L^3$$

Substituting in the second area moment for a square cross section of area t^2

$$I = \frac{t^4}{12}$$

$$\boxed{z_{max} = \frac{12W}{48Et^4}\sqrt{a^2 + b^2}^3}$$

Where:

- $F_B = \frac{1}{2}W = 40 \text{ lbf}$
- $L = \sqrt{a^2 + b^2} = 39.812 \text{ in}$

Stiffness Limiting Case (Using $E = 2.16 \times 10^6 \text{ psi}$ for Hickory 2x4 Wood)

$$z_{max} = \frac{12W}{48Et_1^4} \sqrt{a^2 + b^2}^3 \leq 0.5 \text{ in}$$

$$\frac{12[80 \text{ lbf}]}{48[2.16 \times 10^6 \text{ psi}]t_1^4} [39.812 \text{ in}]^3 \leq 0.5 \text{ in}$$

$$\frac{1}{t_1^4} \leq 0.8557 \text{ in}^{-4}$$

$$\boxed{t_1 \geq 1.0397 \text{ in}}$$

Where:

- $F_B = \frac{1}{2}W = 40 \text{ lbf}$
- $L = \sqrt{a^2 + b^2} = 39.812 \text{ in}$

Strength Limiting Case (Using $\sigma_Y = 1760 \text{ psi}$ for Hickory 2x4 Wood)

$$FS = \frac{\sigma_Y}{\sigma_{all}}$$

To prevent yielding, the allowable load is constrained by $\sigma_{max} \leq \sigma_{all}$. Where the max tensile bending stress is given by:

$$\sigma_{max} = \frac{M_{max}c}{I}$$

Where:

- $M_{max} = M\left(\frac{L}{2}\right) = \frac{F_B L}{2}$
- $C = \frac{t}{2}$
- $I = \frac{t^4}{12}$

$$\sigma_{max} = \frac{\frac{F_B L}{2} \frac{t}{2}}{\frac{t^4}{12}} = \frac{3F_B L}{t^3} \longrightarrow \sigma_{max} \leq \sigma_{all} = \frac{\sigma_Y}{FS} \longrightarrow \frac{3F_B L}{t_2^3} \leq \frac{\sigma_Y}{FS}$$

$$\frac{3[40 \text{ lbf}][39.812 \text{ in}]}{t_2^3} \leq \frac{[1760 \text{ psi}]}{2.5} \longrightarrow \frac{1}{t_2^3} \leq 0.1473 \text{ in}^{-3} \quad \boxed{t_2 \geq 1.8932 \text{ in}}$$

- $F_B = \frac{1}{2}W = 40 \text{ lbf}$
- $L = \sqrt{a^2 + b^2} = 39.812 \text{ in}$
- $FS = 2.5$

Recheck deflection based on strength limiting case

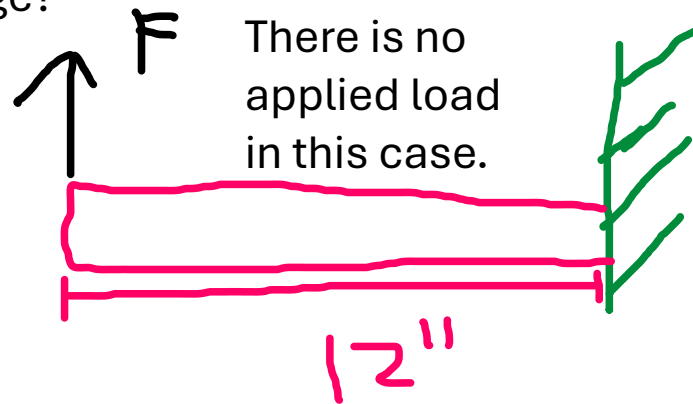
$$\frac{12[80 \text{ lbf}]}{48[2.16 \times 10^6 \text{ psi}][1.9 \text{ in}]^4} [39.812 \text{ in}]^3 = 0.044 \text{ in}$$

Conclusion and assumptions

Given that the strength limiting case was more critical ($t_2 > t_1$), the minimum thickness for the square beam is 1.8932 *in*.

This calculation was done under the assumption that the weight of the person's leg is a point load and is applied only on the diagonal beam. In reality, a continuous plate would apply a distributed loading onto both beams instead of a point load. Additionally, the hinge was modeled to provide zero moment reactions while a real continuous hinge would provide some resistance against twisting and torsion.

Create design models for the beam-column, considering strength and stability (and for now ignoring deflections). The maximum beam length (for now) is 12". What is your beam cross-section? Do you have to consider beam twist/warpage?



$$\sum_x = 0 = F_{\text{Beam}} - F_{\text{Leg}} - F_s$$

$$\sum_{TA} = 0 = -F_{\text{Leg}} \cdot 8.5 + F_s \cdot 17$$

$$8.5 F_{\text{Leg}} = 17 F_s$$

$$F_{\text{Leg}} = 2 F_s$$

$$F_s = \frac{F_{\text{Leg}}}{2}$$

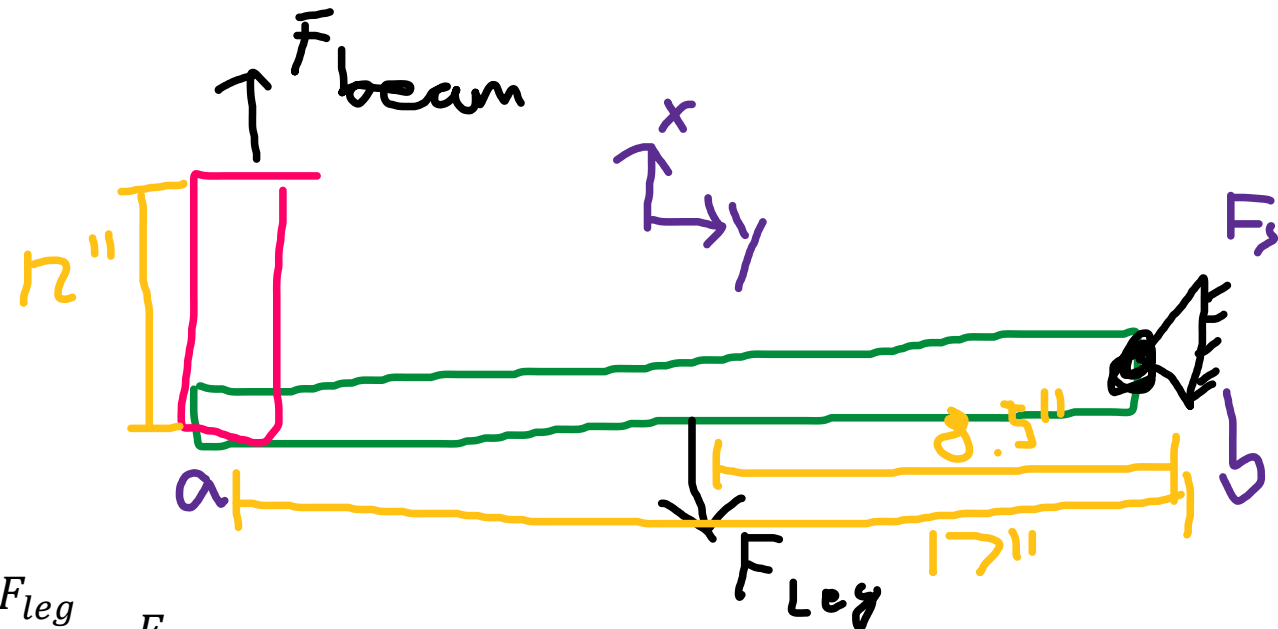
$$F_{\text{leg}} = \frac{F_{\text{leg}}}{2} - F_{\text{beam}}$$

$$F_{\text{beam}} = \frac{F_{\text{leg}}}{2} = 40 \text{ lb}$$

$$E = \frac{\text{stress}}{\text{strain}} = 2.16 \times 10^6 \text{ psi} = \frac{\frac{40 \text{ lb}}{1.9'' \times 1.9''}}{\text{strain}}$$

$$\text{strain} = 5.13 \times 10^{-5}$$

$$FS = \frac{\text{designed}}{\text{expected}} = \frac{a^2 \cdot \sigma_{\text{yield}}}{80} = \frac{1.9 \text{ in}^2 \cdot 1760 \text{ lb/in}^2}{80 \text{ lb}} = 79.42$$



Assume material is hickory wood in tension, which has a E of 2.16×10^6 psi. The most common size is a 2x4 and it is hard to determine the exact yield point of wood so a 2x4 will be used. Beam twisting and warpage is negligible because the Young's modulus multiplied by area is so high, under pure tension the strain is also so low that both are negligible

Choose plate and beam materials and dimensions that meet or exceed the specified requirements in the table, and report the maximum deflection, maximum stresses, and minimum factor of safety according to your design model.

The plate will be made of the beam structure with the beams forming the main structure like previously proposed and a thin plate of arbitrary material filling in the gaps. It will be made of a 1.9" shagbark hickory square wood which is limited by the minimum safety factor of 2.5. With this wood thickness, the deflection is about 0.044 in which is well within the design requirements. The FS for the beam is 79.4 because the beam will be made of the same material as the plate to save manufacturing costs.

Discuss your model limitations, limitations that you put on your design to make the analysis easier, and next steps in the design and analysis process.

The design limitation assumed that there was no deflection coming from the lifting beam. The two edge cases were the leg lift being fully up and down, neither of which caused bending load on the lifting beam. However, this does not matter too much as a 2x4 is a very common and cheap material for building and in tension the strain is in the order of 10^{-6} , meaning that it will deform less than 0.0004" in tension, and will probably not deform too much in bending, seeing as the material is often used in bending for houses. The model then assumes the lifting plate is made up of a two wooden beams with a flexible plate over them. The model also assumes there is no strain in the joints. This is a reasonable model because the two wooden beams forms a triangulated structure that does not rely on fixed supports for the ends. This makes the analysis significantly easier and is a reasonable way to build a lifting plate so this one will be used. The same beams were chosen because it would be most reasonable to simply buy a large quantity of similar beams and cut them to length than to source difference different sized beams and join them together.

Sources cited

- <https://amesweb.info/Materials/Youngs-Modulus-of-Wood.aspx>
- Beer, F. P., Johnston, E. R., Jr., DeWolf, J. T., & Mazurek, D. F. (2012). *Mechanics of materials* (6th ed.). McGraw-Hill.
- <https://www.woodworkweb.com/woodwork-topics/wood/146-wood-strengths.html>